

ORM (Object Relational Mapping) in Odoo

ORM is one of the **most important concepts** in Odoo. Every Odoo developer uses it daily.

What is ORM?

ORM = Object Relational Mapping

It is a technology that lets you **communicate with the database using Python code instead of writing SQL queries**.

In simple words:

ORM acts as a translator between Python and the PostgreSQL database.

Python Code



ORM



SQL Query



PostgreSQL Database

Without ORM

Suppose you want to see all customers.

You would write SQL:

```
SELECT * FROM res_partner;
```

Here you need to know:

- SQL syntax

- Table names
 - Database structure
-

With ORM

In Odoo, you simply write Python:

```
partners = self.env['res.partner'].search([])
```

The ORM automatically converts it into SQL behind the scenes.

Python



ORM converts



```
SELECT * FROM res_partner;
```

You never see the SQL—it is generated automatically.

Real-Life Example

Imagine you go to a restaurant.

Without ORM

You go directly to the kitchen and tell the chef what to cook.

Customer



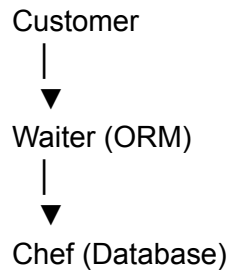
Chef

This is like writing SQL yourself.

With ORM

You tell the waiter what you want.

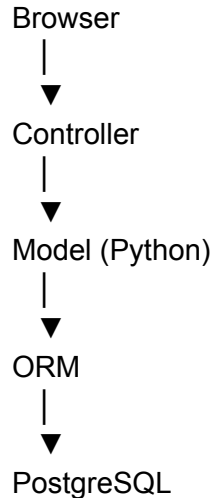
The waiter talks to the chef and brings your food.



The **waiter** is like the ORM.

- You talk in normal language (Python).
 - The waiter translates it.
 - The chef (database) prepares the result.
-

ORM in Odoo Architecture



The ORM is the bridge between your Python models and the database.

CRUD Operations with ORM

CRUD means:

Operation	Meaning	ORM Method
C	Create	<code>create()</code>
R	Read	<code>search()</code> , <code>browse()</code>
U	Update	<code>write()</code>
D	Delete	<code>unlink()</code>

1. Create

Create a new student:

```
self.env['student.student'].create({  
    'name': 'Sugumar',  
    'age': 48  
})
```

ORM converts it to something like:

```
INSERT INTO student_student (name, age)  
VALUES ('Sugumar', 48);
```

2. Read

Read all students:

```
students = self.env['student.student'].search([])
```

ORM converts it to:

```
SELECT * FROM student_student;
```

3. Update

Update a student:

```
student.write({
```

```
'age': 49  
})
```

ORM converts it to:

```
UPDATE student_student  
SET age = 49;
```

4. Delete

Delete a student:

```
student.unlink()
```

ORM converts it to:

```
DELETE FROM student_student;
```

Why Odoo Uses ORM

1. Easier Development

Instead of SQL:

```
SELECT * FROM res_partner;
```

You write:

```
self.env['res.partner'].search([])
```

2. More Secure

ORM helps protect against **SQL Injection** attacks by safely handling database queries.

3. Less Code

Python code is shorter and easier to understand.

4. Database Independence

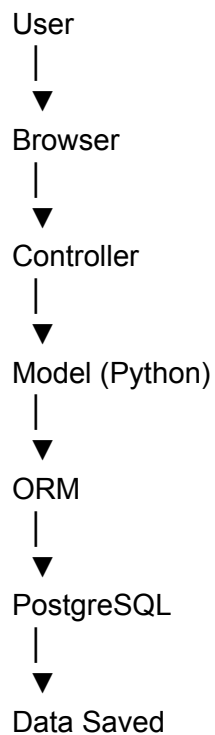
Your code focuses on business logic rather than database-specific SQL syntax.

5. Automatic Validation

ORM handles many checks and integrates with Odoo's security and business rules.

ORM Flow Example

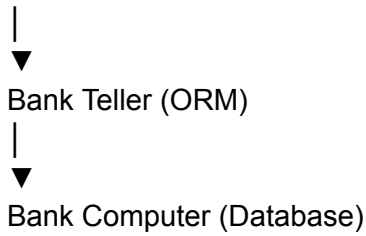
Suppose you click **Save** after entering a new customer.



Simple Analogy

Imagine a bank.

You



You don't access the bank's computer directly.

You speak to the teller.

The teller performs the transaction for you.

Similarly:

- **Python = You**
 - **ORM = Bank Teller**
 - **PostgreSQL = Bank Database**
-

Advantages of ORM

- ☒ No need to write SQL for common operations.
 - ☒ Easier to learn and maintain.
 - ☒ Integrates with Odoo's security model.
 - ☒ Reduces development time.
 - ☒ Produces cleaner, more readable code.
-

Interview Questions

Q1. What is ORM?

Answer: ORM (Object Relational Mapping) is a technology that allows developers to interact with the database using Python objects and methods instead of writing SQL queries directly.

Q2. Which database does Odoo ORM communicate with?

Answer: PostgreSQL.

Q3. Name the four main ORM CRUD methods.

Answer:

- `create()` – Create records
 - `search()` – Read/Search records
 - `write()` – Update records
 - `unlink()` – Delete records
-

One-Line Definition to Remember

ORM (Object Relational Mapping) is the bridge between Python and the PostgreSQL database. It automatically converts Python code into SQL queries, allowing developers to work with database records without writing SQL directly.